

$$S = k \ln \Omega, \quad \frac{1}{\Omega} = P_i$$

$$\left\{ \begin{array}{l} P_i = \frac{1}{Z} e^{-\beta E_i} \\ \langle E \rangle = - \frac{\partial}{\partial \beta} \ln Z \end{array} \right. \Rightarrow Z = \sum_i e^{-\beta E_i}; \beta = \frac{1}{k_B T}$$

مثال: گاز ایدئال تک اتمی: $(q_1, q_2, \dots, q_{3N}, p_1, p_2, \dots, p_{3N})$: کمالات

$$H = \sum_{i=1}^{3N} \frac{p_i^2}{2m} \Rightarrow Z = \frac{1}{h^{3N} N!} \int_{-\infty}^{+\infty} Dq Dp e^{-\beta H}$$

$$Z = \frac{1}{h^{3N} N!} \int_{-\infty}^{+\infty} dq_1 \dots dq_{3N} dp_1 \dots dp_{3N} e^{-\beta \left(\frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \dots + \frac{p_{3N}^2}{2m} \right)}$$

$$Z = \frac{V^N}{h^{3N} N!} \left(\int_{-\infty}^{+\infty} dp_1 e^{-\beta \frac{p_1^2}{2m}} \right)^{3N} = \frac{V^N}{N! h^{3N}} \underbrace{\left(\sqrt{2m\pi} \beta^{-1/2} \right)^{3N}}_{\sqrt{2m\pi kT}}$$

$$\frac{h}{\sqrt{2m\pi kT}} = \lambda_{th} \Rightarrow \left(\frac{V}{\lambda_{th}^3} \right)^N \frac{1}{N!}$$

$$\ln Z = \ln \left(\frac{V^N (2m\pi)^{3N}}{h^{3N} N!} \right) - \frac{3N}{2} \ln \beta$$

$$U = \langle E \rangle = - \frac{\partial}{\partial \beta} \ln Z = \frac{3N}{2} \frac{1}{\beta} = \boxed{\frac{3}{2} N k T}$$

$$C_v = \left(\frac{\partial U}{\partial T} \right)_v = \frac{3}{2} N k_B = \frac{3}{2} n R \Rightarrow \frac{3}{2} R$$

ندادرات ↑ $N_A k_B$

(α, β)

$$H = H(\alpha) + H(\beta)$$

$$Z = \int e^{-\beta H} d\alpha d\beta = \int e^{-\beta H(\alpha)} d\alpha \int e^{-\beta H(\beta)} d\beta$$

$$= Z_1(\alpha) Z_2(\beta)$$

$$Z = \text{tr} e^{-\beta \hat{H}}$$

$$C_p - C_v = T \left(\frac{\partial S}{\partial T} \right)_p - T \left(\frac{\partial S}{\partial T} \right)_v \quad S(T, p) = S(T, p(T, v))$$

$$= S(T, v)$$

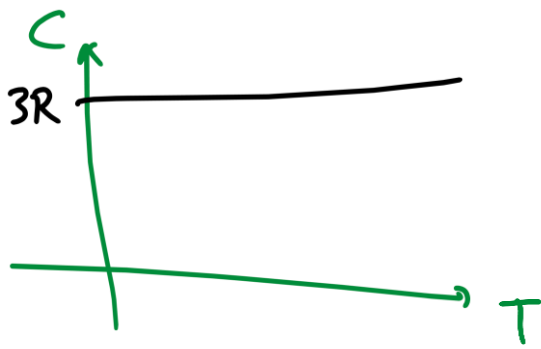
$$\left(\frac{\partial S}{\partial T} \right)_v = \left(\frac{\partial S}{\partial T} \right)_p + \left(\frac{\partial S}{\partial p} \right)_T \left(\frac{\partial p}{\partial T} \right)_v$$

$$\frac{C_p - C_v}{T} = \cancel{\left(\frac{\partial S}{\partial T} \right)_p} - \cancel{\left(\frac{\partial S}{\partial T} \right)_p} - \underline{\left(\frac{\partial S}{\partial p} \right)_T} \underline{\left(\frac{\partial p}{\partial T} \right)_v}$$

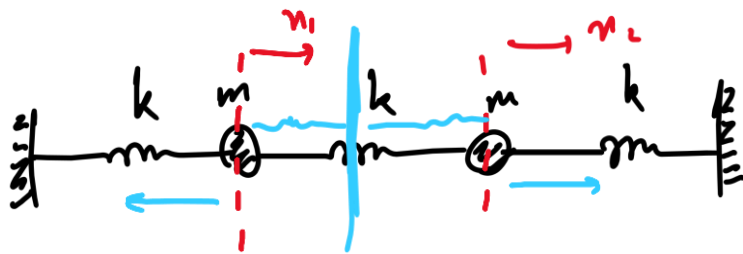
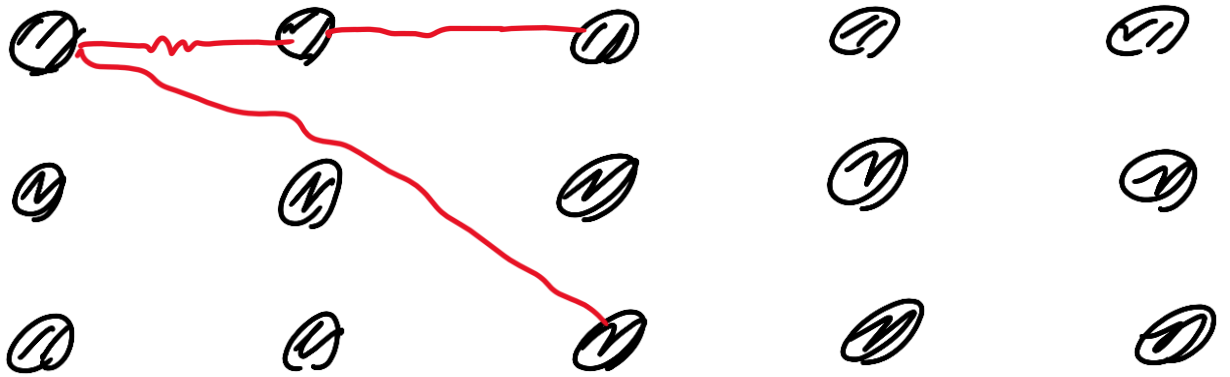
$$\left(\frac{\partial S}{\partial p} \right)_T = - \left(\frac{\partial V}{\partial p} \right)_T \Rightarrow \frac{C_p - C_v}{T} = \underline{\left(\frac{\partial V}{\partial T} \right)_p} \underline{\left(\frac{\partial p}{\partial T} \right)_v}$$

$\propto$

$$\left(\frac{\partial p}{\partial T} \right)_v = - \frac{\frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p}{\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T} = \frac{\alpha}{\kappa_T} \Rightarrow C_p - C_v = VT \frac{\alpha^2}{\kappa_T}$$



$$\boxed{\frac{C}{R} = 0,735}$$



$$(n_1, n_2, \dot{n}_1, \dot{n}_2)$$

$$L = \frac{1}{2} m (\dot{n}_1^2 + \dot{n}_2^2) - \frac{1}{2} k (n_1^2 + n_2^2 + (n_1 - n_2)^2)$$

$$P_1 = \frac{\partial L}{\partial \dot{n}_1} = m \dot{n}_1, \quad P_2 = m \dot{n}_2$$

$$H = \frac{P_1^2}{2m} + \frac{P_2^2}{2m} + \frac{1}{2} k (n_1^2 + n_2^2 + (n_1 - n_2)^2)$$

$$\frac{\partial L}{\partial q_i} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} : \begin{cases} m \ddot{n}_1 = -k n_1 - k (n_1 - n_2) \\ m \ddot{n}_2 = -k n_2 + k (n_1 - n_2) \end{cases} \quad \sqrt{\frac{k}{m}} = \omega$$

$$\ddot{n}_1 + \omega^2 (2n_1 - n_2) = 0$$

$$\ddot{n}_2 + \omega^2 (2n_2 - n_1) = 0$$

$$\begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} A \\ B \end{pmatrix} e^{i\alpha t} \implies \begin{aligned} n_1 &= A e^{i\alpha t} \\ n_2 &= B e^{i\alpha t} \end{aligned}$$

$$-\alpha^2 A + \omega^2 (2A - B) = 0$$

$$-\alpha^2 B + \omega^2 (2B - A) = 0$$

$$\begin{pmatrix} -\alpha^2 + 2\omega^2 & -\omega^2 \\ -\omega^2 & -\alpha^2 + 2\omega^2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{vmatrix} -\alpha^2 + 2\omega^2 & -\omega^2 \\ -\omega^2 & -\alpha^2 + 2\omega^2 \end{vmatrix} = 0 \implies \underline{\alpha = \pm \omega, \pm \sqrt{3}\omega}$$

$$\begin{pmatrix} \omega^2 & -\omega^2 \\ \omega^2 & \omega^2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies A - B = 0$$

$$\xrightarrow{\alpha = \omega} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \longleftarrow$$

$$\begin{pmatrix} -\omega^2 & -\omega^2 \\ -\omega^2 & -\omega^2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies A = -B$$

$$\omega = \sqrt{3} \omega$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\begin{cases} X_1 = \frac{1}{\sqrt{2}} (n_1 + n_2) & , & P_1 = \frac{p_1 + p_2}{\sqrt{2}} \end{cases}$$

$$\begin{cases} X_2 = \frac{1}{\sqrt{2}} (n_1 - n_2) & , & P_2 = \frac{p_1 - p_2}{\sqrt{2}} \end{cases}$$

$$n_1 = \frac{1}{\sqrt{2}} (X_1 + X_2) \quad n_2 = \frac{1}{\sqrt{2}} (X_1 - X_2)$$

$$\{n_i, n_j\} = \{p_i, p_j\} = 0 \quad , \quad \{n_i, p_j\} = \delta_{ij}$$

$$\longrightarrow \{X_i, X_j\} = \{P_i, P_j\} = 0 \quad , \quad \{X_i, P_j\} = \delta_{ij}$$

$$H = \frac{P_1^2}{2m} + \frac{P_2^2}{2m} + \frac{1}{2} k (n_1^2 + n_2^2 + (n_1 - n_2)^2)$$

$$H = \frac{1}{2m} \left(\frac{P_1 + P_2}{\sqrt{2}} \right)^2 + \frac{1}{2m} \left(\frac{P_1 - P_2}{\sqrt{2}} \right)^2 + \frac{1}{2} k \left(\left(\frac{X_1 + X_2}{\sqrt{2}} \right)^2 + \left(\frac{X_1 - X_2}{\sqrt{2}} \right)^2 + (\sqrt{2} X_2)^2 \right)$$

$$\frac{P_1^2}{2m} + \frac{P_2^2}{2m} + \frac{1}{2} k (X_1^2 + X_2^2 + 2X_2^2)$$

$$\omega_2 := \sqrt{\frac{3k}{m}}$$

$$\omega_1 = \sqrt{k/m}$$

$$H = \underbrace{\frac{P_1^2}{2m} + \frac{1}{2} m \omega_1^2 X_1^2}_{\omega_1} + \underbrace{\frac{P_2^2}{2m} + \frac{1}{2} m \omega_2^2 X_2^2}_{\omega_2}$$

$$Z = \int dn_1 dn_2 dP_1 dP_2 e^{-\beta H}$$

$$X_1 = \frac{1}{\sqrt{2}} (n_1 + n_2) \quad X_2 = \frac{1}{\sqrt{2}} (n_1 - n_2)$$

$$\begin{vmatrix} \frac{\partial X_1}{\partial n_1} & \frac{\partial X_1}{\partial n_2} \\ \frac{\partial X_2}{\partial n_1} & \frac{\partial X_2}{\partial n_2} \end{vmatrix} = \begin{vmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{vmatrix} = -1$$

$$H = \frac{1}{2m} P^2 + \frac{1}{2} m \omega^2 n^2$$

$$Z = \int dn dP e^{-\beta \left(\frac{1}{2m} P^2 + \frac{1}{2} m \omega^2 n^2 \right)}$$



$$\sqrt{\frac{2\pi m}{\beta}} \sqrt{\frac{2\pi}{m \omega^2 \beta}} = \frac{2\pi}{\omega} \beta^{-1}$$

$$\ln Z = \ln \frac{2\pi}{\omega} - \ln \beta \implies \langle E \rangle = \frac{1}{\beta} = \underline{k_B T}$$

$$Z = \int dx_1 dx_2 dp_1 dp_2 e^{-\beta H}$$

$$= \int dx_1 dx_2 dp_1 dp_2 e^{-\beta \left(\frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \frac{1}{2} m \omega_1^2 x_1^2 + \frac{1}{2} m \omega_2^2 x_2^2 \right)}$$

$$= \sqrt{\frac{2\pi m}{\beta}} \sqrt{\frac{2\pi m}{\beta}} \sqrt{\frac{2\pi}{m \omega_1^2 \beta}} \sqrt{\frac{2\pi}{m \omega_2^2 \beta}} = \frac{(2\pi)^2}{\omega_1 \omega_2 \beta^2}$$

$$= \frac{(2\pi)^2}{\omega_1 \omega_2} \beta^{-2} \longrightarrow Z = \frac{(2\pi)^N}{\omega_1 \omega_2 \dots \omega_N} \beta^{-N}$$

$$\langle E \rangle = 2 k_B T$$

$$\langle E \rangle = N k_B T$$

$$H = \frac{p^T p}{2m} + \frac{1}{2} X^T \underline{k} X$$

$$U U^T = I$$

$$\tilde{k} = U k U^T \implies U^T \tilde{k} U = k$$

$$H = \frac{\underbrace{p^T}_{\tilde{p}^T} \underbrace{U^T U}_{I} p}{2m} + \frac{1}{2} X^T \underbrace{U^T}_{\tilde{X}^T} \tilde{k} \underbrace{U X}_{\tilde{X}}$$

$$H(q_1, q_2, \dots, q_N) = \alpha q_1^2 + H'(q_1, \dots, q_N)$$

$$Z = \frac{1}{h^{3N}} \int dq_1 Dq e^{-\beta H'} e^{-\beta \alpha q_1^2}$$

$$\sqrt{\frac{\pi}{\beta \alpha}} Z' \Rightarrow \ln Z = \ln C - \frac{1}{2} \ln \beta + \ln Z'$$

$$\Rightarrow \langle E \rangle = U' + \frac{1}{2} k_B T$$

$$(q_1, q_2, \dots, q_{3N}, p_1, p_2, \dots, p_{3N})$$

$$H = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \dots + \frac{p_{3N}^2}{2m} + \frac{1}{2} m \omega_1^2 X_1^2 + \frac{1}{2} m \omega_2^2 X_2^2 + \dots + \frac{1}{2} m \omega_{3N}^2 X_{3N}^2$$

$$U = \langle E \rangle = 6N \frac{k_B T}{2} = 3N k_B T$$

$$C = \left(\frac{\partial U}{\partial T} \right)_V = 3N k_B \Rightarrow C_{\text{per atom}} = 3k_B$$

$$E_n = \hbar \omega \left(n + \frac{1}{2} \right)$$

$$\rightarrow H = \frac{p^T p}{2m} + \frac{1}{2} q^T \underline{k} q + \underline{a^T q}$$

$$H = \frac{p^2}{2m} + \overbrace{a q^2 + \underline{b q}}$$

$$P = p, \quad Q = q + k^{-1} a$$